

ERC Starting Grant 2026

Part B1

Flexible Intelligence through Neurocomputational Design

FIND

Principal Investigator: Dr. Mikkel Elle Lepperød
Host Institution: Simula Research Laboratory, Norway
Duration: 60 months
Primary Panel: PE6 — Computer Science and Informatics

Abstract

In the next decade, AI must make the leap to human-like cognitive flexibility—requiring lasting memory, causal understanding, and continuous adaptation. Current AI systems remain constrained by bounded computational frameworks that struggle with long-term memory, causal reasoning, and catastrophic forgetting. FIND proposes a paradigm shift through **PAM** (**P**redictive **A**ctive temporal coding with distributed **M**emory), a novel framework where modules of neurons read and write to local long-term memory, drawing inspiration from biological memory systems and adhering to the “shallow brain hypothesis”.

PAM combines three key mechanisms into a unified architecture: (1) *Distributed modular specialization* where each module maintains dedicated memory space for independent information processing; (2) *Temporal predictive coding* where continuous predictions and error-driven updates govern memory and learning; and (3) *Active inference* where action selection minimizes uncertainty about predictions. Our preliminary implementation already achieves 98.1% exact recall at 40% input noise and $3.75\times$ less forgetting than ablated baselines.

FIND will develop PAM through three work packages: **WP1** establishes the temporal predictive coding architecture with distributed memory for processing information over different time scales; **WP2** extends this to causal reasoning and subspace-based compositional understanding through neurosymbolic integration; and **WP3** stress-tests the complete system in demanding open-ended environments (mazes, Minecraft, Factorio) requiring sophisticated planning, resource management, and long-horizon adaptation. FIND’s long-term goal is to develop AI systems capable of the brain’s memory capacity for exploration, reasoning, and adaptive knowledge transfer.

FIND: Flexible Intelligence through Neurocomputational Design

1 Part I of the Scientific Proposal

We live in time. Memory serves as the essential bridge connecting our past to our present, enabling an agent to maintain continuity of experience, learn from previous encounters, and make informed decisions. Humans can develop internal models of how the world works, enabling planning and reasoning by simulating potential outcomes. Replicating this ability in machines might be the next step toward flexible AI agents [1]. Compared to humans, current AI systems struggle to generalize their knowledge to new situations [2]. AI is limited by the lack of long-term memory [3], causal understanding [4], and the ability to adapt through life-long learning without catastrophic forgetting [5]. The prevailing technologies—recurrent neural networks, state space models, and attention-based models—remain constrained by their bounded computational frameworks [3, 6]. In scenarios where it is necessary to understand the effects of past events, stable long-term memory is essential. This raises the question: **How can systems intelligently select what information to retain or discard, and what to retrieve, in order to learn across long temporal sequences?**

1.1 Ground-breaking nature and impact

FIND proposes a paradigm shift in how neural networks are built, drawing inspiration from biological memory systems. We develop **PAM** (**P**redictive **A**ctive temporal coding with distributed **M**emory), a novel framework where modules of neurons are capable of reading and writing to local long-term memory. This framework adheres to the “shallow brain hypothesis” [7] and operates through:

- Specialized neural modules with dedicated memory systems [8, 9, 10]
- Symbolic representations [11] retained in memory
- Learned orchestration mechanisms that dynamically route information based on task demands
- Probabilistic processing of inputs and actions to generate temporal predictions
- Action selection that minimizes uncertainty [12, 13, 14, 15]

FIND’s novelty lies at the intersection of three established fields in a configuration no prior system has achieved simultaneously:

- **Temporal predictive coding** [14] with **distributed external memory**—not just recurrent state: prior tPC models store context solely in hidden activations; FIND externalises memory into content-addressable HDC slots persistent across thousands of timesteps.
- **Memory-augmented networks** [16] with **biologically-grounded iterative inference**—not single-pass read/write: prior MANNs use a single forward pass to address memory; FIND uses iterative predictive-coding inference to resolve ambiguous queries, yielding a $1.75\times$ recall advantage at $\sigma=0.4$.
- **Hyperdimensional computing** [11] **embedded as the storage format of a gradient-trained system**—not as a standalone symbolic method: prior HDC systems operate outside neural training loops; FIND integrates HDC binding directly into the differentiable memory update, enabling end-to-end learning.

1.2 State of the art, knowledge needs, and project objectives

Temporal memory and predictive coding

A key challenge in AI is creating models with memory capabilities comparable to the brain. Recent advances have unified in-context learning with both retrieval-augmented generation (RAG) [17, 18] and reinforcement learning, creating more adaptive AI systems. However, these remain palliative solutions that sidestep the fundamental challenge of bottom-up lifelong learning, as they depend on human-curated memory augmentation and struggle with out-of-distribution scenarios [2]. **These failures highlight the gap between current AI systems and truly autonomous learning agents that can continuously adapt through direct experience.**

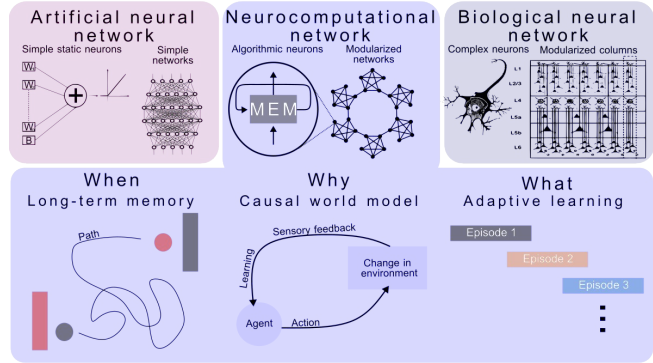


Figure 1: Conceptual overview of FIND. We develop neurocomputational networks whose modules hold local long-term memories, striking an algorithmic balance between overly simple ANNs and overly complex biological neural networks. Computations are orchestrated across specialised modules to address three core challenges: *long-term memory* (recalling past experiences during decision making); *causal reasoning* (learning cause–effect relationships through active environmental interventions); and *adaptive learning* (acquiring new knowledge continuously without forgetting).

The core challenge for long-term memory in neural networks is separating memory storage from computation [19, 20]. Memory-augmented neural networks (MANNs)—including NTMs [21], DNCs [16], and Kanerva Machines [8, 10]—have begun to address this. Hyperdimensional computing (HDC) extends sparse distributed memories by enabling symbolic operations through vector binding and composition [11, 22]. **Despite these advances, significant challenges remain in developing memory systems that bridge neural and symbolic representations while compressing temporal information.**

Predictive coding (PC) offers a unified framework by minimizing prediction errors between top-down expectations and bottom-up inputs [13, 12]. Recent extensions to active [12, 15] and temporal PC [14, 23] enable biologically plausible cognitive maps. **However, these models rely on Markov assumptions, making them unable to efficiently handle long-term memory and credit assignment.**

Causal reasoning and world models

A causal world model represents the generative mechanisms by which actions produce effects—enabling counterfactual reasoning and robust generalisation [24, 4]. Recent developments include JEPA [25] and DINO-WM [26], while S4 world models show promise in model-based RL [6]. However, the time horizons in current benchmarks are irrelevant for real-world problems: 2000-step benchmarks [6] do not reflect the capabilities of desert ants, which maintain accurate memories after 200 000 steps [27]. **Transformers and S4 face critical limitations in long-term memory due to their limited context windows [28].**

Latent learning of cognitive maps has been pursued in neuroscience [29, 30] and AI [31, 32], including by our lab [33, 34, 35, 36], but these approaches lack a unified theory of perception, cognition, and action—the gap FIND addresses through memory-augmented temporal predictive coding.

Adaptive learning

Adaptable memory for life-long learning requires systems to dynamically manage information without catastrophic forgetting. Current approaches span regularization, replay buffers, and Bayesian variational methods [5, 37, 38]; in our lab we have explored generative approaches to both stability [39] and plasticity [40]. FIND addresses this through hyperdimensional computing [11, 22]: by binding memories as compositional high-dimensional vectors, the system enables both fast approximate retrieval and explicit symbolic queries, supporting life-long learning without retraining from scratch.

1.3 Research strategy and objectives

Main hypothesis: Modular specialization can emerge naturally from predictive principles and provide long-term memory support for active inference and adaptive learning, enabling causal world models without catastrophic forgetting.

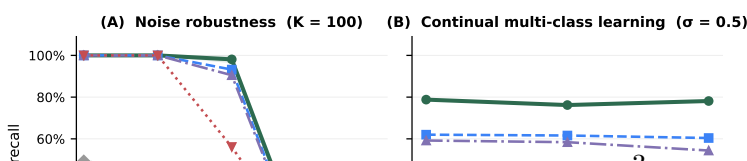
We address this through three objectives and corresponding work packages:

- **Objective 1 / WP1: Temporal Predictive Coding with Distributed Memory.** Develop the foundational PAM architecture that processes information over different time scales through temporal predictive coding with distributed memory modules.
- **Objective 2 / WP2: Causal Reasoning and Compositional Understanding.** Extend PAM with active inference, neurosymbolic integration, and subspace-based compositional reasoning for causal world model learning.
- **Objective 3 / WP3: Long-Horizon Adaptation in Open-Ended Environments.** Stress-test the complete PAM system in demanding environments (mazes, Minecraft, Factorio) that require sophisticated planning, resource management, and long-horizon adaptation.

Theoretical approach: PAM combines three mechanisms into a unified architecture: (1) *Distributed modular specialization*: PAM distributes memory across multiple specialized modules, each with dedicated memory space inspired by Kanerva machines [8]. (2) *Temporal predictive coding*: The system generates predictions about future sensory inputs; prediction errors drive memory and model updates. (3) *Active inference*: The system actively chooses actions to minimize uncertainty about its predictions, enabling efficient exploration and causally relevant memory storage.

PAM is based on content-addressable memory [8, 16] with retrieval via similarity between the current latent state z_t^m and stored memory basis vectors h_i^m . We use exponential maps [41] that respect the curved geometry of high-dimensional hypervectors, enabling closed-form Riemannian updates. HDC [11, 41] bridges neural and symbolic approaches: implementing sparse distributed memories where symbolic relationships are encoded as high-dimensional vectors, and enabling compositional reasoning across modules through binding operations.

Preliminary results. The BioAI group at Simula has developed the four-



dations of FIND through a coherent line of research on neurocomputational memory and orchestration. In computational neuroscience, we have established that AI agents spontaneously develop grid, place, and border cells [33], and that orthogonal remapping encodes context-dependent memory [35, 36, 34]. In neural orchestration, we have demonstrated that Neural Cellular Automata (NCA) can serve as self-organizing inter-module communication mechanisms [42], that NCA-based engram networks support one-shot memory transfer [43], and that NCA produce developmental solutions to abstract reasoning tasks [44].

Building directly on these results, we have implemented and benchmarked the outline of the PAM architecture: the tPC-KM system achieves 98.1% exact recall at 40% input noise versus 56.1% for a feedforward encoder—a $1.75\times$ gap attributable solely to iterative inference. Under online concept drift, the system forgets $3.75\times$ less than the tPC-alone baseline (Figure 2).

1.4 Major expected outcomes

Science: A new theoretical framework unifying predictive coding, distributed memory, and active inference that transcends the Markov assumptions of current models. Fundamental understanding of how modular specialization can emerge naturally from predictive principles.

Technology: The PAM architecture as an open-source framework for memory-augmented temporal reasoning, with comprehensive benchmarks. A new class of orchestration algorithms for inter-module communication.

Society: Foundations for AI systems with principled memory and causal reasoning capabilities applicable to healthcare (patient timeline reasoning), industrial automation (long-horizon production planning), and environmental monitoring (long-range sensor fusion).

References

- [1] Y. LeCun. <https://openreview.net/pdf?id=BZ5a1r-kVsf>. 2022.
- [2] Y. Razeghi et al. (2022). arXiv: 2202.07206.
- [3] J. Pasukonis et al. (2022). arXiv: 2210.13383.
- [4] J. Richens et al. (2024). arXiv: 2402.10877.
- [5] L. Wang et al. (2024). DOI: 10.1109/tpami.2024.3367329.
- [6] F. Deng et al. (2023). arXiv: 2307.02064.
- [7] M. Suzuki et al. (2023). DOI: 10.1038/s41583-023-00756-z.
- [8] Y. Wu et al. 2018. arXiv: 1804.01756.
- [9] P. Kanerva. 2003. ISBN: 978-0-262-11132-4.
- [10] A. H. Marblestone et al. (2020). arXiv: 2002.02385.
- [11] P. Kanerva (2009). DOI: 10.1007/s12559-009-9009-8.
- [12] K. Friston (2010). DOI: 10.1038/nrn2787.
- [13] B. Millidge et al. 2022. DOI: 10.24963/ijcai.2022/774.
- [14] B. Millidge et al. (2024). DOI: 10.1371/journal.pcbi.1011183.
- [15] R. P. N. Rao et al. (2023). ISSN: 1530-888X. DOI: 10.1162/neco_a_01627.
- [16] A. Graves et al. (2016). DOI: 10.1038/nature20101.
- [17] P. Lewis et al. (2020).
- [18] C. Zhang et al. (2023). arXiv: 2309.01390.
- [19] M. Gemici et al. (2017). arXiv: 1702.04649.
- [20] C. R. Gallistel et al. 2011. ISBN: 978-1-405-12288-7.
- [21] A. Graves et al. (2014). arXiv: 1410.5401.
- [22] D. Ma et al. (2022). arXiv: 2207.12932.
- [23] J. Gornet et al. (2024). DOI: 10.1038/s42256-024-00863-1.
- [24] B. Scholkopf et al. (2021). DOI: 10.1109/jproc.2021.3058954.
- [25] K. Drozdov et al. (2024). arXiv: 2412.10925.
- [26] G. Zhou et al. (2024). arXiv: 2411.04983.
- [27] Z. Hu et al. (2024). arXiv: 2402.07465.
- [28] T. Ni et al. (2023). arXiv: 2307.03864.
- [29] T. Hafting et al. (2005). DOI: 10.1038/nature03721.
- [30] J. O’Keefe (1976). DOI: 10.1016/0006-8993(76)90358-7.
- [31] J. C. Whittington et al. (2020). DOI: 10.1016/j.cell.2020.10.024.
- [32] A. Banino et al. (2018). DOI: 10.1038/s41586-018-0102-6.
- [33] V. Schøyen et al. (2022). ISSN: 1556-5068. DOI: 10.2139/ssrn.4309105.
- [34] M. B. Pettersen et al. (2025). DOI: 10.7554/elife.99302.
- [35] M. Lepperød et al. 2024. DOI: 10.52202/079017-0008.
- [36] V. S. Schøyen et al. (2025). ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1012804.
- [37] J. Nassar et al. (2022).
- [38] C. V. Nguyen et al. (2017). arXiv: 1710.10628.
- [39] M. Wasiluk et al. <https://openreview.net/pdf?id=2LhCPowI6i>. 2023. arXiv: 2309.01895.
- [40] A. Singh et al. (2025). arXiv: 2503.04728.
- [41] M. Pettersen et al. (2025). DOI: 10.1101/2025.09.25.678504. eprint: <https://www.biorxiv.org/content/early/2025/09/25/2025.09.25.678504.full.pdf>.
- [42] M.-K. Kvalsund et al. (2026). DOI: 10.1016/j.neunet.2026.108798.
- [43] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.831.
- [44] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.833.

2 Curriculum Vitae and Track Record

Personal details

Name: Dr. Mikkel Elle Lepperød
Date of birth: 14 September 1985
Nationality: Norwegian
ORCID: 0000-0002-4262-5549
URL: <https://www.simula.no/people/mikkel>

Education and key qualifications

2020 **Ph.D.** Neuroscience, Faculty of Medicine, University of Oslo, Norway
 2014 **M.Sc.** Applied Mathematics, Norwegian University of Life Sciences, Norway

Current and relevant previous positions

2026– **Senior Research Scientist** (equiv. associate professor), Simula Research Laboratory
 2023– **Adjunct Associate Professor** (20%), Dept. of Physics, University of Oslo
 2022–2026 **Research Scientist** (equiv. assistant professor), Simula Research Laboratory
 2020–2022 **Postdoctoral Fellow**, Dept. of Biosciences, University of Oslo

PhD supervisor: Prof. Torkel Hafting and Prof. Marianne Fyhn (University of Oslo).

Postdoctoral mentor: Self-directed (group leader from 2020).

Career breaks: Parental leave (5 months in 2017, 5 months in 2020).

Research outputs

The following ten outputs demonstrate my trajectory from experimental neuroscience to neuro-inspired AI, establishing the foundations for FIND:

1. **Lepperød ME** et al. *Optogenetic pacing of medial septum parvalbumin-positive cells disrupts temporal but not spatial firing in grid cells*. Science Advances, 2021. Role: First author, conceived and led the study. Significance: Established causal evidence for temporal coding in the entorhinal cortex, directly motivating the temporal predictive coding component of PAM.
2. **Lepperød ME** et al. *Inferring causal connectivity from pairwise recordings and optogenetics*. PLoS Computational Biology, 2022. (With K. Kording.) Role: First author. Significance: Developed quasi-experimental methods for causal inference in neural circuits, grounding FIND's causal reasoning objectives.
3. Schøyen VS*, Pettersen MB*, . . . , **Lepperød ME**. *Coherently remapping toroidal cells but not grid cells are responsible for path integration in virtual agents*. iScience, 2023. Role: Senior/-corresponding author, supervised the project. Significance: Showed that AI agents spontaneously develop biologically consistent spatial cells, establishing our lab's cognitive map expertise.
4. Pettersen MB, Schøyen VS, Maltre-Sørenssen A, **Lepperød ME**. *Decoding the cognitive map: Learning place cells and remapping*. eLife, 2024. Role: Senior/corresponding author. Significance: Demonstrated context-dependent memory through orthogonal remapping, key to FIND's modular memory design.
5. Pettersen MB, Rogge F, **Lepperød ME**. *Learning place cell representations and context-dependent remapping*. NeurIPS, 2024. Role: Senior author. Significance: Extended cognitive map learning to context-switching, directly informing WP2's compositional reasoning.
6. Kvalsund MK, Ellefsen KO, Glette K, Pontes-Filho S, **Lepperød ME**. *Sensor movement drives emergent attention and scalability in active neural cellular automata*. Neural Networks, 2026. Role: Senior/corresponding author. Significance: Demonstrated NCA-based inter-module communication with active vision—the primary orchestration mechanism for PAM.
7. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *EngramNCA: A neural cellular automaton model of memory transfer*. ALife, 2025. Role: Co-supervisor. Significance: Showed NCA-based one-shot memory transfer between modules, directly enabling PAM's distributed memory architecture.
8. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *ARC-NCA: Towards developmental solutions to the Abstraction and Reasoning Corpus*. ALife, 2025. Featured on Machine Learning Street Talk podcast. Role: Co-supervisor. Significance: Demonstrated abstract reasoning through developmental self-organization, informing WP2's compositional reasoning.

9. Schøyen VS et al. *Hexagons all the way down: Grid cells as a conformal isometric map of space*. PLOS Computational Biology, 2025. *Role: Senior/corresponding author. Significance: Provided geometric theory of neural representations, informing HDC integration in PAM.*
10. Danielli K, **Lepperød ME**. *A bio-inspired minimal model for non-stationary K-armed bandits*. Biological Cybernetics, 2026. *Role: Senior/corresponding author. Significance: Demonstrated bio-inspired adaptive learning under non-stationarity, directly relevant to WP3's open-ended adaptation.*

Significant peer recognition

- **NORA Award for Early Career Investigator** (2025)—National recognition for contributions to NeuroAI
- **Simula ERC Incubator** (2022)—Selected for institutional support program for ERC candidates
- **Wharton Executive Education** (2022)—“High-Potential Leaders: Accelerating Your Impact”
- **Invited speaker:** University of California San Diego (2022), AI Week Oslo Met (2023), Keynote at Young Researchers in NeuroAI (2023), Simula Bio-inspired Computing Seminar (2024)
- **Workshop organizer:** 4-day NeuroAI workshop on Hurtigruten with 20 international speakers (2024); co-organizer of 3+ NeuroAI workshops connecting national and international labs (2023–2024)
- **Reviewer:** Nature Machine Intelligence, PLoS Computational Biology, eNeuro
- **Board member:** NORA Special Interest Group for NeuroAI Norway
- **Public outreach:** Brain Inspired podcast (2024), Machine Learning Street Talk podcast (2025), NRK interview (2021), opinion pieces in digi.no and forskning.no

Project leadership and supervision

Group leader, BioAI group at Simula (2020–present): 4.5 MNOK/year, managing 3 PhD students, 1 postdoc. Built the group from inception to a productive research unit with publications at NeurIPS, eLife, Science Advances, and Nature Communications.

Acting co-PI, Bio-inspired neural networks for AI application (2020–2026): 16 MNOK RCN-funded project, University of Oslo. Co-supervised 3 PhDs.

Supervised to completion: 2 PhD students (V. Schøyen, K. Holzhausen), 8+ MSc students (including graduates now at Stanford PhD program and UiO PhD program).

Additional information

Contributions to research infrastructure: Co-developed NEST simulator modules (Human Brain Project); co-developed Exdir data format and Expipes data management platform; lead role establishing NeuroAI as focus field in Norway–US Ministry of Education collaboration (2022). Developed NeuroAI course at University of Oslo (2023).

3 Part II of the Scientific Proposal

This section details the research methodology, work plan, and risk assessment for FIND. Part II complements Part I by providing the implementation specifics; it should not be read as a repetition of Part I but as a detailed elaboration of how the objectives will be achieved.

Research questions:

- Q1: How can memory-augmented modules distribute and combine long-term temporal information?
- Q2: How can such modules support active inference and causal reasoning?
- Q3: How can neurosymbolic components be integrated into distributed memory architectures without sacrificing gradient-based learning?
- Q4: Does this integration measurably enhance compositional reasoning in complex environments?
- Q5: Can such networks learn adaptively and reason about causal relations over long time horizons?

3.1 Research methodology

The core update equations underpin PAM’s mechanisms. Each module m has latent state $z_t^m \in \mathbb{R}^{d_z}$ and sensory input $x_t^m \in \mathbb{R}^{d_x}$ at time t . The prediction error $e_t = x_t^m - \hat{x}_t^m$ drives module state updates: $z_{t+1}^m = f_m(z_t^m, c_t^m, r_t^m, a_t)$, where f_m is a learnable function, a_t is the action, $r_c^m = \sum_i \alpha_c^m[i] h_i^m$ is context-dependent memory retrieval, and c_t^m is contextual input from inter-module communication. Two primary orchestration mechanisms will be explored:

1. **Message passing:** attention and convolution combined with aggregation—demonstrated in active vision [1]
2. **Neural cellular automata (NCA):** self-organizing principles for module communication—demonstrated in one-shot memory transfer [2] and abstract reasoning [3]

Four further mechanisms form a structured ablation ladder in WP2: *Hypernetwork*, *Predictive routing*, *Competitive collaboration*, and *Energy-based orchestration*—ordered by increasing biological plausibility and implementation complexity.

WP1: Temporal Predictive Coding with Distributed Memory (Lepperød, PhD1; Q1; Years 1–3)

WP1 establishes the core PAM architecture for temporal prediction and memory. Building on our preliminary results (98.1% exact recall at $\sigma=0.4$, $3.75\times$ less forgetting than tPC-alone), WP1 extends the architecture to challenging temporal benchmarks and characterizes its operational regime.

T1.1: Temporal Benchmarks and Architecture Scaling

Completed: MNIST digit recall benchmark [4] (see Part I, Figure 2). **Planned:** Extend to Omniglot [5], “infinite dSprites” [6], Memory Maze offline probing [7], and Minecraft Malmo [8]. Visual encoding via ResNet [9]. Compare against GTMs [4], VRNNs [10], LSTMs [11], S4 [12], NTMs [13], and DNCs [14]. Offline probing [15] will analyze learned representations. *Success criterion:* PM must achieve $>90\%$ exact recall at $\sigma=0.3$ and above-chance performance across all benchmark phases.

T1.2: Inference Depth, Noise Robustness, and Capacity

Systematically characterise the recall–noise–capacity landscape across $\sigma \in [0.0, 0.8]$ and $K \in [10, 500]$. Implement adaptive inference depth via prediction-error thresholding. Characterise interactions between slot count, HDC dimension, and forgetting rate under varying concept-drift speeds, providing design guidelines for WP2–3. *Success criterion:* Adaptive inference must achieve $\geq 95\%$ of full-iteration accuracy at $\leq 60\%$ of compute on the MNIST benchmark.

T1.3: Hierarchical Temporal Compression for Long-Horizon Memory

Extend PM with hierarchical time-scale factorisation: fast modules ($\Delta t=1$) process step-level inputs while slow modules compress episodes into semantic summaries via learned gating inspired by complementary learning systems [16]. The gating criterion—whether and when prediction error triggers consolidation from episodic to semantic memory—will be resolved experimentally using prediction-error statistics from T1.2. Evaluate on “infinite dSprites” and 2,000–10,000 step sequences in Memory Maze [7].

WP2: Causal Reasoning and Compositional Understanding (Lepperød, PD; Q2–Q4; Years 2–4)

WP2 extends PAM with active inference, causal world model learning, and subspace-based compositional reasoning. This WP integrates neurosymbolic components for explicit reasoning alongside statistical learning, while enabling the system to discover causal structure through interventions.

T2.1: Action-Augmented Temporal Predictive Coding

Extend PM to incorporate action nodes that modulate state transitions. Develop action-dependent prediction error computation across multiple timescales, including action-conditioned memory ad-

dressing, action-dependent update rules, and methods for maintaining temporal consistency. Design joint optimization of state predictions and action selections. *Success criterion:* Action-conditioned prediction error must correlate $r > 0.7$ with observed state transitions in a controlled navigation task.

T2.2: Subspace Reasoning and Compositional Understanding

Develop mechanisms for compositional, subspace-based reasoning within the PAM architecture. Implement methods for factorizing latent spaces into semantically meaningful subspaces, enabling compositional manipulation of concepts through HDC binding and unbinding operations [17, 18]. The modular architecture enables different modules to represent different subspaces, supporting analogical reasoning across domains. *Success criterion:* Compositional retrieval accuracy must exceed $>80\%$ on novel concept combinations not seen during training.

T2.3: Causal World Model Learning through Active Inference

Implement active inference for action selection balancing information gain and goal-oriented reward [19, 20]. Develop causal discovery mechanisms through interventionist exploration: modules learn not just correlations but causal direction through the consequences of actions. Build on our work with NCA [21, 1] and proximal policy optimization [22] for credit assignment. *Success criterion:* PAM with active-inference exploration must reach task completion $2\times$ faster than standard PPO on at least two benchmark environments.

T2.4: Orchestration Ablation and Evaluation

Execute comprehensive ablation studies to quantify the contribution of each orchestration mechanism, distributed memory, and neurosymbolic component. Evaluate PAM on few-shot continuous control [23, 24] and long-horizon planning [7]. Investigate how PAM’s computations map onto neural circuits by extending our established RNN framework [25, 26, 27] that demonstrates spontaneous development of grid, place, and border cells. *Success criterion:* PAM must achieve >10 percentage point improvement over LSTM baseline on Memory Maze ($T > 500$ steps). Orchestration ablations must show primary NCA/message-passing design outperforms at least two alternatives.

WP3: Long-Horizon Adaptation in Open-Ended Environments (Lepperød, PhD2; Q4–Q5; Years 3–5)

WP3 stress-tests PAM in environments demanding sophisticated planning, resource management, and adaptation to ever-changing dynamics. We evaluate in Minecraft [28] and Factorio [29]: Factorio tests long-term memory through complex production chains; Minecraft challenges spatial recall and creative problem-solving.

T3.1: Environment Development and Curriculum Design

Design complex scenarios with varied challenges: resource management, crafting, building, combat, and exploration. Create a procedural generation pipeline varying task complexity, resource constraints, and causal dependencies. Implement learning progress-based curriculum using automatic curriculum learning [30] and intrinsic motivation [31]. Integration via Python Malmö API [8] for Minecraft and Factorio RCON interface; observations via frozen ResNet [9] encoder and game-state variables.

T3.2: Long-Horizon Active Inference Agent

Integrate PAM into an interactive agent with a three-level policy hierarchy: (L1) step-level motor primitives via PPO [22]; (L2) sub-goal selection via hindsight experience replay [32]; (L3) strategic planning as symbolic goal vectors in HDC space [17, 18]. Co-train through RL and self-supervised prediction objectives.

T3.3: Causal and Adaptive Learning Analysis

Assess causal learning through interventions (changing game rules, introducing disturbances), counterfactual reasoning (simulating alternative scenarios), and analyzing the agent’s ability to predict effects of its actions. Measure module specialization by analyzing activity patterns, stored information types, and task-specific contributions. Compare against DQN [33], PPO, RND [34], Dreamer [35], DINO-WM [36], and MuZero [37]. *Success criteria:* (i) PAM achieves >10 pp improvement over DQN on at least two Minecraft sub-tasks after 10^6 steps. (ii) In Factorio, PAM reaches production throughput $\geq 1.5\times$ that of DreamerV3 on the automation-tier task. (iii) Intervention experiments demonstrate $>20\%$ accuracy in predicting novel action effects.

3.2 Work and time schedule

Team allocation: Lepperød (PI, 50%) provides scientific leadership across all WPs and leads WP2. The Postdoc (PD, 100%) joins in Year 1 and co-leads WP2

with Lepperød, contributing to T2.1–T2.4. PhD1 (100%) starts Year 1 and leads WP1 (T1.1–T1.3). PhD2 (100%) starts Year 2 and leads WP3 (T3.1–T3.3). An additional PhD student may be hired in Year 3 depending on progress.

3.3 Feasibility, risks, and contingency plans

Cat.	Description	Like.	Sev.	Mitigation
Scientific	Modular architecture fails to specialize.	Med	High	Progressive training, soft inductive biases, architectural constraints.
	Memory does not overcome catastrophic forgetting.	Med	High	Hybrid memory combining fast/slow learning; Bayesian consolidation.
	tPC-KM fails to scale beyond MNIST.	Med	Med	Start with simplified environments; adaptive memory pruning.

Cat.	Description	Like.	Sev.	Mitigation
Technical	Computational resources exceed expectations.	Med	Med	Sigma2 allocation + Simula eX3 cluster; model distillation.
	Orchestration bottlenecks in information flow.	Med	High	Structured ablation ladder provides fallback mechanisms.
Impl.	Environment integration (Minecraft/Factorio).	Med	Med	Simplified environment wrappers with fallback to Memory Maze.
Project	Recruitment of specialized expertise.	Med	Med	Leverage NORA network, international collaborations.

Feasibility: The preliminary results (98.1% recall, $3.75\times$ less forgetting) demonstrate that the

core architecture works. Each WP has independent value: even if WP3’s most ambitious targets are not fully reached, WP1–2 produce publishable architectures and benchmarks. The structured ablation ladder in WP2 ensures that at least one orchestration mechanism will succeed.

3.4 Collaborations

Prof. Konrad Kording (University of Pennsylvania): Bayesian decision theory and causal inference (WP2). One co-authored publication, ongoing collaboration since 2018.

Prof. Marc de Kamps (University of Leeds): Computational neuroscience and compositional learning (WP2–3). Two co-authored publications, collaboration since 2016.

Assoc. Prof. Blake Richards (Mila/McGill): Normative models of hippocampal activity. Co-PI, PhD exchange program, ongoing since 2023.

Assoc. Prof. Arvind Kumar (KTH): Neuromodulators in AI. Co-PI and co-author, ongoing since 2020.

Prof. Stefan Leutgeb (UC San Diego): Experimental spatial navigation. Visiting researcher 2017/2018, co-authored work submitted.

3.5 Integration into host institution

Simula Research Laboratory is a Norwegian research institution with a strong track record in computational science. FIND will be hosted in the Department of Numerical Analysis and Scientific Computing (SCAN), within the BioAI group led by Lepperød. Simula provides:

- **Compute infrastructure:** Access to Simula’s eX3 HPC cluster and national Sigma2 (Saga, Olivia) resources. WP3 RL experiments are estimated to require $\sim 10^5$ GPU-hours.
- **Research environment:** Active AI research community, including the SURE-AI centre for industry collaboration, and connections to University of Oslo through Lepperød’s adjunct professorship.
- **Training programs:** Simula Academy for researcher development; NORA Research School for PhD training; established supervision track record (2 PhDs completed, 4 in progress).

3.6 Recruitment and ethics

Recruitment: PhD and postdoc positions will be advertised internationally through NORA, ELLIS, and Simula’s networks. We actively seek diverse candidates and will leverage our international collaborations for candidate identification. Simula’s competitive salaries and Oslo’s quality of life support strong recruitment.

Ethics: This research focuses on fundamental methods development in controlled game environments. No human subjects, animal data, or personal data are involved. We will follow Simula’s Code of Ethics, the EU Ethics Guidelines for Trustworthy AI, and the EU AI Act. Beyond the project, applications of FIND technology would require evaluation of biases, security, and data governance.

Funding ID

Funder	Amount	Subject	Status
Simula internal	4.5 MNOK/yr	BioAI group baseline funding	Active
RCN IKTPLUSS	12 MNOK	PASSION: AI safety and alignment	Submitted
RCN FRIPRO	10 MNOK	FIND (FRIPRO version): NeuroAI memory	Not funded
RCN IKTPLUSS	12 MNOK	constrAIIn: Constrained AI methods	Submitted

The proposed ERC project is distinct from all listed funding. The FRIPRO FIND application (3-year, smaller scope) was not funded; the ERC proposal represents a substantially expanded 5-year programme with additional WPs, team members, and ambitious open-ended environment testing not included in the FRIPRO version.

References

- [1] M.-K. Kvalsund et al. (2026). DOI: 10.1016/j.neunet.2026.108798. [2] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.831. [3] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.833. [4] M. Gemici et al. (2017). arXiv: 1702.04649. [5] B. M. Lake et al. (2015). DOI: 10.1126/science.aab3050. [6] S. Dziadzio et al. (2023). arXiv: 2312.16731. [7] J. Pasukonis et al. (2022). arXiv: 2210.13383. [8] M. Johnson et al. 2016. DOI: 10.24963/ijcai.2016/643. [9] K. He et al. 2016. DOI: 10.1109/cvpr.2016.90. [10] J. Chung et al. (2015). arXiv: 1506.02216. [11] S. Hochreiter et al. (1997). DOI: 10.1162/neco.1997.9.8.1735. [12] A. Gu et al. (2021). arXiv: 2111.00396. [13] A. Graves et al. (2014). arXiv: 1410.5401. [14] A. Graves et al. (2016). DOI: 10.1038/nature20101. [15] K. Park et al. (2023). arXiv: 2311.03658. [16] J. Ramapuram et al. (2021). arXiv: 2103.03905. [17] P. Kanerva (2009). DOI: 10.1007/s12559-009-9009-8. [18] B. Aragam et al. 2024. DOI: 10.52202/079017-3211. [19] K. Friston (2010). DOI: 10.1038/nrn2787. [20] R. P. N. Rao et al. (2023). ISSN: 1530-888X. DOI: 10.1162/neco_a_01627. [21] M. A. Alsakkal et al. (2025). arXiv: 2501.18799. [22] J. Schulman et al. (2017). arXiv: 1707.06347. [23] A. Mete et al. 2024. arXiv: 2402.00158. [24] T. Yu et al. (2019). arXiv: 1910.10897. [25] V. Schøyen et al. (2022). ISSN: 1556-5068. DOI: 10.2139/ssrn.4309105. [26] M. B. Pettersen et al. (2025). DOI: 10.7554/elife.99302. [27] M. Lepperød et al. 2024. DOI: 10.52202/079017-0008. [28]

W. Hill et al. (2023). arXiv: 2312.12891. [29] N. Kant (2025). arXiv: 2502.01492. [30] R. Portelas et al. (2020). arXiv: 2003.04664. [31] A. Aubret et al. (2023). DOI: 10.3390/e25020327. [32] M. Andrychowicz et al. 2017. arXiv: 1707.01495. [33] V. Mnih et al. (2015). DOI: 10.1038/nature14236. [34] Y. Burda et al. (2018). arXiv: 1808.04355. [35] D. Hafner et al. (2018). arXiv: 1811.04551. [36] G. Zhou et al. (2024). arXiv: 2411.04983. [37] J. Schrittwieser et al. (2019). arXiv: 1911.08265.